

Recall double integrals $\iint f(x,y) dx dy$, $dx dy$ being tiny flat squares across 2D space [xy plane is completely flat].

If the weighted func. $f(x,y) = 1$, the double integral simply calculates the flat area of the region.

If the weighted func. $f(x,y) \neq 1$, the double integral calculates the volume above the flat area or a weighted flat area.

Double integrals are entirely confined to flat, 2D planes.

If we need to calc. properties along a curved surface in 3D space, we need tiny curved squares, dS not $dy dx$.

The action of summing up these curved squares is known as surface integrals.

Double int. cuz Area $\iint f(x,y,z) dS$ tiny curved squares

However, $\iint$ only can handle 2 variables, but we have 3, (x,y,z) .

And also, for a curved surface, x,y,z are dependent on each other [e.g. moving a step in x in a sphere, def increases z], thus impossible to integrate partially.

$\therefore$ To solve this issue, we parameterize the 3D curved surface as a flat 2D map. The parameterization func. $\vec{r}(u,v)$ maps every flat coordinate to its specific 3D location (x,y,z)

As expected the tiny square on the flat uv plane will be distorted when projected onto the 3D space. Thus to compensate, we multiply the flat tiny 2D square, $du dv$ with the rate of change of the area $\left[\frac{d\vec{r}}{du} \times \frac{d\vec{r}}{dv} \right]$. Thus $dS = \left| \frac{d\vec{r}}{du} \times \frac{d\vec{r}}{dv} \right| du dv$
Recall area = $\vec{a} \times \vec{b}$

Surface Integral = $\iint r(u,v) \left| \frac{d\vec{r}}{du} \times \frac{d\vec{r}}{dv} \right| du dv$
weighted func. compensated tiny curved squares

Flux

Flux is just the ammount of $\vec{F}(x,y,z)$ vector field going perpendicularly through a dS surface.

$$\iint \vec{F} \cdot \vec{n} \, dS$$

unit normal vector

weighted func.

tiny curve square

Further simplifying.

$$\iint \left[\vec{F}(x,y,z) \cdot \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} \right] |\vec{r}_u \times \vec{r}_v| \, du \, dv$$

$$\iint \vec{F}(\vec{r}(u,v)) \cdot (\vec{r}_u \times \vec{r}_v) \, du \, dv$$

Surface Eqn
[to get the $\vec{F}$ at
the wanted surface]

$$\iint \vec{F}(\vec{r}(u,v)) \cdot \vec{N} \, du \, dv$$

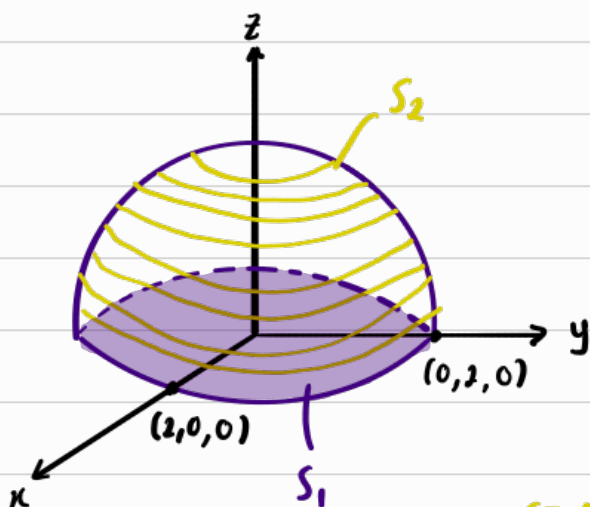
Take Note :

For an open surface, we can choose the inward/outward normal vector

For a closed surface [e.g. sphere], the convention is to choose outward pointing normal. [A +ve flux means the net force is going outwards]

Example

Let S be the surface enclosing the upper half ball of radius 2 and centre at the origin. Find the flux across the surface S given the vector field $\vec{F} = \langle 7x, 0, -(z+1) \rangle$



Flux through S_1

Polar coordinates
treating $u = r, v = \theta$
 $\vec{r}(r, \theta) = \vec{r}(u, v)$

$$\vec{r}(u, v) = \langle u \cos v, u \sin v, 0 \rangle$$

for $0 \leq u \leq 2, 0 \leq v \leq 2\pi$

$$\begin{aligned} \vec{N} &= \frac{d\vec{r}}{du} \times \frac{d\vec{r}}{dv} \\ &= \begin{bmatrix} \cos v \\ \sin v \\ 0 \end{bmatrix} \times \begin{bmatrix} u(-\sin v) \\ u \cos v \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0 \\ u(\cos^2 v + \sin^2 v) \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0 \\ u \end{bmatrix} \quad \text{inward [in to shape]} \end{aligned}$$

for $0 \leq u \leq 2$

outward [out from shape]

$\vec{N}$ pointing outward is $-\begin{bmatrix} 0 \\ 0 \\ u \end{bmatrix}$ # convention

$$\begin{aligned} \phi_{S_1} &= \int_0^{2\pi} \int_0^2 \begin{pmatrix} 7x \\ 0 \\ -(z+1) \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ -u \end{pmatrix} du dv \\ &= \int_0^{2\pi} \int_0^2 u(z+1) du dv \quad \text{z=0 in } S_1 \\ &= \int_0^{2\pi} \int_0^2 u du dv \\ &= 4\pi \quad \# \end{aligned}$$

Flux through S_2

Spherical coordinate
Since $r = 2$, let $u = \theta, v = \phi$
 $\vec{r}(u, v) = \vec{r}(u, v)$

$$\vec{r}(u, v) = \langle 2 \cos u \sin v, 2 \sin u \sin v, 2 \cos v \rangle$$

for $0 \leq u \leq 2\pi, 0 \leq v \leq \frac{\pi}{2}$

$$\vec{N} = \begin{bmatrix} -2 \sin u \sin v \\ 2 \cos u \sin v \\ 0 \end{bmatrix} \times \begin{bmatrix} 2 \cos u \cos v \\ 2 \sin u \cos v \\ -2 \sin v \end{bmatrix}$$

$$= -4 \langle \cos u \sin^2 v, \sin u \sin^2 v, \sin v \cos v \rangle$$

check direc.

Let $u = 0, v = \frac{\pi}{2}$

$$\vec{N}(0, \frac{\pi}{2}) = -4 \langle 1, 0, 0 \rangle = \begin{pmatrix} -4 \\ 0 \\ 0 \end{pmatrix}$$

inward (outward)

$\therefore \vec{N}$ pointing outward is $4 \begin{bmatrix} \cos u \sin^2 v \\ \sin u \sin^2 v \\ \sin v \cos v \end{bmatrix}$

$$\begin{aligned} \therefore \vec{F} \cdot \vec{N} &= \begin{pmatrix} 7x \\ 0 \\ -(z+1) \end{pmatrix} \cdot 4 \begin{pmatrix} \cos u \sin^2 v \\ \sin u \sin^2 v \\ \sin v \cos v \end{pmatrix} \\ &= \begin{pmatrix} 7(2 \cos u \sin v) \\ 0 \\ -(2 \cos v + 1) \end{pmatrix} \cdot 4 \begin{pmatrix} \cos u \sin^2 v \\ \sin u \sin^2 v \\ \sin v \cos v \end{pmatrix} \\ &= 56 \sin^3 v \cos^2 u - 8 \cos^2 v \sin v - 4 \sin v \cos v \end{aligned}$$

$$\begin{aligned} \phi_2 &= \int_0^{\frac{\pi}{2}} \int_0^{2\pi} (56 \sin^3 v \cos^2 u - 8 \cos^2 v \sin v - 4 \sin v \cos v) du dv \\ &= 28\pi \end{aligned}$$

$$\begin{aligned} \therefore \phi_{\text{total}} &= \phi_1 + \phi_2 \\ &= 4\pi + 28\pi = 32\pi \quad \# \end{aligned}$$

